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Crystallization behavior and microstructure of powdered and bulk $ZnO–Al_2O_3–SiO_2$ glass-ceramics

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Abstract

The crystallization behavior of glass with the composition: 55.6 mol% SiO_2, 22.8 mol% Al_2O_3, 17.7 mol% ZnO and 3.84 mol% of TiO_2 as nucleating agent and with different particle sizes has been studied by differential scanning calorimetry (DSC), X-ray diffraction (XRD) and tranmission electron microscopy (TEM). In glass powders two crystalline phases: zinc–aluminosilicate s.s. with high-quartz structure, $Zn_{x/2}Al_xSi_{3-x}O_6$, (x varies dependent on heat-treatment temperature) and gahnite are formed. The ratio of these phases depends on particle sizes. In bulk glass, however, gahnite is the sole crystalline phase. The composition of initially formed zinc–aluminosilicate s.s. was determined by Rietveld refinement of XRD patterns to be $Zn_{0.69}Al_{1.38}Si_{1.62}O_6$. With temperature increase, the amount of zinc–aluminosilicate s.s decreased with simultaneous reduce of zinc and aluminum incorporated in the structure. Eventually at 1423 K almost pure high-quartz structure was formed. The activation energies of zinc–aluminosilicate s.s. and gahnite crystallization were determined by non-isothermal method to be 510 ± 18 and 344 ± 17 kJ mol^{-1}, respectively. The latter value matches well with those cited in literature for crystal growth of gahnite in similar glasses. That is attributed to the fact that the high-quartz structure acts as a precursor for gahnite crystallization.

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1. Introduction

Transparent glass-ceramics based on spinel compositions with gahnite ($ZnAl_2O_4$) as the main crystalline phase can be prepared from glasses on the basis of SiO_2–Al_2O_3–ZnO system, with ZrO_2 and/or TiO_2 as nucleating agents [1–6]. According to Beall and Pickney [3] nucleation of glasses with TiO_2 or with both nucleating agents is preceded by a highly uniform, ultra-fine-scale phase separation into SiO_2-rich and TiO_2/Al_2O_3-rich areas. Gahnite ($ZnAl_2O_4$) then crystallizes and grows within the latter globular areas [3,4]. Studies conducted with powdered glasses [7] show that gahnite arises by two simultaneous mechanisms: growing on nuclei present in the whole glass, and from zinc–aluminosilicate solid solution with high-quartz structure that crystallizes on the surface of glass particles and acts as a precursor for gahnite. Which mechanism prevails depends on glass particle size. These glass-ceramics have excellent thermal stability. Potential applications of these materials include those where transparency and high use temperature are important, including flat panel displays and certain photovoltaic substrates [3]. Glazes containing gahnite are also interesting, since it is the crystalline phase of high scratch hardness and high refractive index [8].

In this work, the influence of particle size on phase development in $ZnO–Al_2O_3–SiO_2$ glasses with TiO_2 as nucleating agent has been studied. The composition of zinc–aluminosilicate s.s. in dependence of heat treatment temperature was studied by Rietveld refinement method

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[9]. The activation energy for crystallization of zinc–aluminosilicate s.s (also called zinc-stuffed derivative of high-quartz) and gahnite was determined under non-isothermal conditions using Kissinger method [10].

2. Experimental procedure

2.1. Preparation of parent glass

The parent glass was prepared from reagent grade chemicals: SiO_2, $Al(OH)_3$, ZnO, and TiO_2. The batch was melted in Pt crucible in electrical furnace for 2h at 1600 °C. Glass was crushed and remelted for another 2h at the same temperature. Remelted melt was poured onto a heated steel plate to form a round discs about 5mm thick. The glass was then annealed at about 925K, and cooled down at the cooling rate of the furnace. The composition of the parent glass is shown in Table 1. The as-prepared glass was ground and sieved to produce 5 different fractions of glass powders (<63, 63–125, 125–250, 250–500 μm and ~5 mm) suitable for further analyses.

2.2. Characterization methods

Characteristic temperatures such as glass transition, T_g, onset of crystallization T_o and crystallization maximum T_p were determined by Netzsch STA 409 Simultaneous Thermal Analyser used in DSC mode. To investigate the crystallization mechanism of zinc–aluminosilicate and gahnite, the DSC scans were performed at the heating rates of 5, 10, 15, 20 and 25 K min^{-1}. The activation energies for crystallization of both phases (zinc–aluminosilicate s.s. and gahnite) under non-isothermal condition, E_a, were calculated from the relationship between the heating rate, β, and the temperature of the peak maximum, T_p, on DSC scans using Kissinger equation:

$$\ln\frac{\beta}{T_p^2} = -\frac{E_a}{RT_p} + \text{const.} \quad (1)$$

where R is the universal gas constant.

The thermally treated samples were analyzed by XRD to identify the crystalline phases formed during DSC runs. X-ray diffraction analysis was carried out on a computer-controlled diffractometer (Siemens D500/PSD) using CuKα radiation with α-quartz single-crystal monochromator, and a curved position sensitive detector. Samples were dispersed on a Si single crystal holder. Data were collected between 5 and 70° 2θ, with steps of 0.02° and counting time of 3 s per step. For the purpose of the Rietveld structure refinement [9], the XRD data were collected using a Philips X'Pert diffractometer with CuKα radiation. XRD patterns were scanned in steps of 0.02° (2θ) in the range from 15 to 120° with a fixed counting time of 30 s per step. The composition of metastable zinc–aluminosilicate s.s. differs from that cited in JCPDS-ICDD file No. 34-1455; therefore, the Rietveld refinement (TOPAS2 program [11]) was performed to determine its composition. The refinement of zinc–aluminosilicate structure was started using $P6_222$ space group [12] and the structure parameters for $LiAlSi_2O_6$ derived by Mueller et al. [13], since zinc prefers the tetrahedral coordination sites in silicates [14] and there is no data for zinc stuffed derivative structure of high-quartz in literature. The refinement for gahnite was started using Fm-3m space group derived by O'Neill [15]. The pseudo Voigt function was used for the modeling of diffraction profiles. In the final refinements the following parameters were refined: scale factor, 2θ zero, seven coefficients of Chebyshev polynomial to describe the background, peak profile parameters, unit cell parameters determined using Si as an standard, x, y, z atom positions, occupation factors

Table 1
Glass composition determined by EDX analysis

	SiO_2	Al_2O_3	ZnO	TiO_2
mol%	55.61	22.83	17.75	3.84

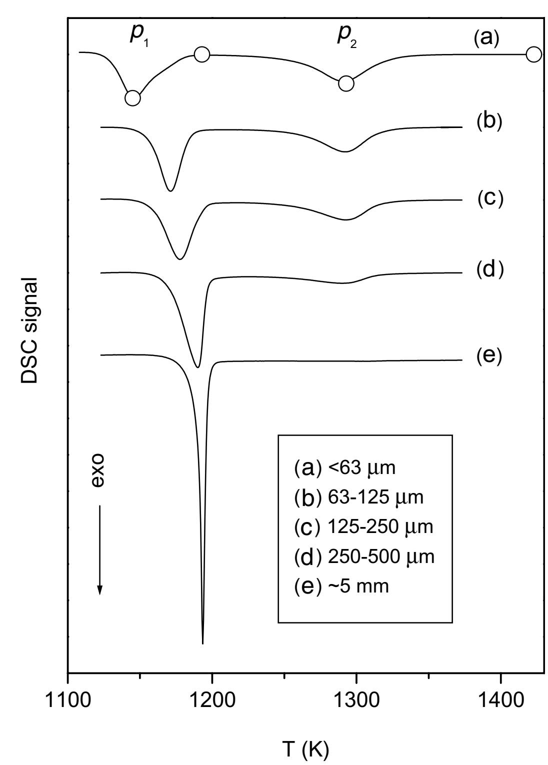

Fig. 1. DSC scans for non-isothermal crystallization of glasses as a function of particle size, (heating rate 10 K min^{-1}). (a) glass particles <63 μm; (b) glass particles 63–125 μm; (c) glass particles 125–250 μm; (d) glass particles 250–500 μm, (e) glass particles ~5 mm. Scans are shifted for visualization purposes. Circles on DSC scan (a) assign the quench temperatures for subsequent XRD analyses.

and isotropic temperature factors. The samples contain some amount of glassy phase, therefore for quantitative evolution of phases, CaF_2 as an internal standard was added in amount of 10wt%, and the Rietveld refinement was repeated using the previously determined unit cell parameters and x, y, z positions of atoms in each of the present structures. Only scale factor, background parameters and 2θ zero shift were refined. Each structural model was refined to convergence, with the best result selected on the basis of reliability factors and stability of the refinement.

Microstructure and microchemical composition of the bulk glass and attained glass-ceramics were characterized by transmission electron microscopy equipped with an EDX detector (TEM, Jeol, JEM 2011 operating at 200kV). The samples for observation were obtained by means of conventional techniques, involving mechanical polishing, dimple grinding and Ar^+ ion beam etching. EDX analysis was also performed to find the exact composition of the parent-glass. The disc of the parent-glass was analyzed at 10 different positions and the mean composition is given in Table 1.

3. Results

DSC scans of a glass with the composition given in Table 1, and with different particle size are shown in Fig. 1. DSC scan of bulk sample (~5mm) exhibits only one sharp exothermic peak at 1193K, (scan (e)), however, on the scans of glass powders (scans (a)–(d)) there exist two separate peaks, p_1 and p_2. The position and intensity of the peak p_1 depends on particle size; as the particle size increases, the peak also increases and shifts to higher temperatures. On the other hand, there is no shift of the second peak, p_2, only its height decreases with the increase of the particle size. XRD analyses taken at various stages of DSC traces were performed to identify the exothermic events observed on the scans in Fig. 1. XRD patterns of the samples quenched at

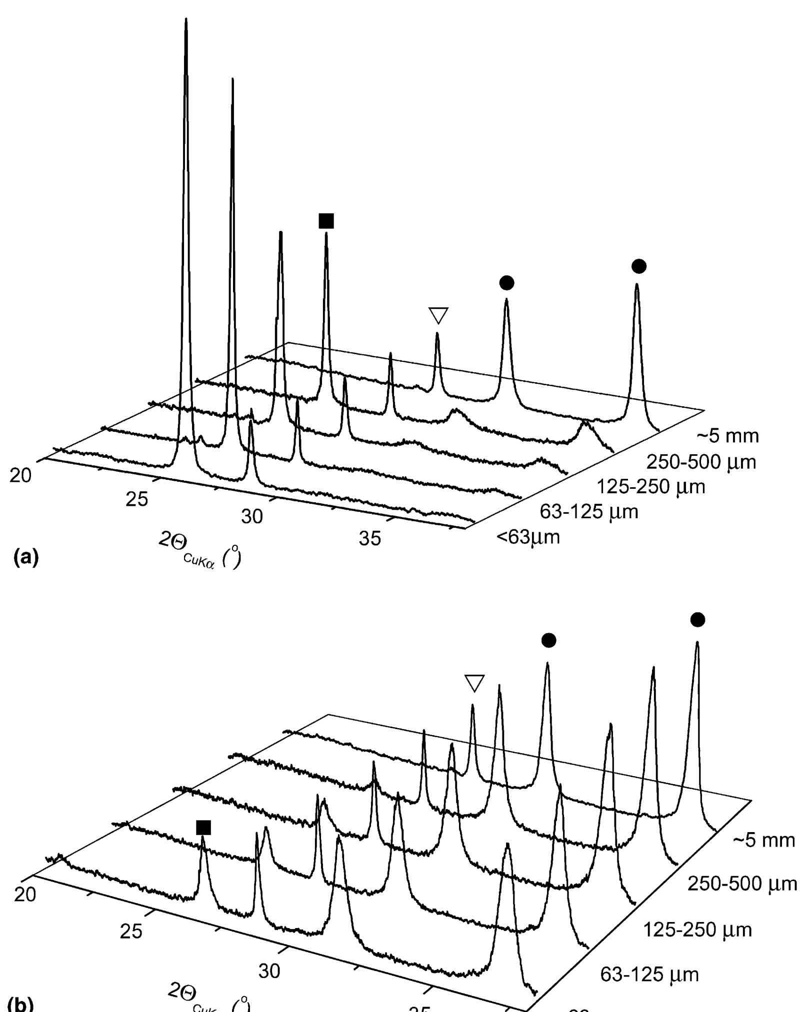

Fig. 2. XRD patterns of glasses with different particles size quenched: (a) at 1193K; (b) at 1423K. Particle sizes are marked on y axes. -■- $Zn_{x/2}Al_xS_{3-x}O_6$, -●- $ZnAl_2O_4$, and -▽- fluorite (added as internal standard).

1193 K and at 1423 K are shown in Fig. 2. (Fluorite, CaF_2, was added as an internal standard). As seen, gahnite is the sole phase determined in the bulk glass at 1193 K and at 1423 K. On the contrary, in glass powders two crystalline phases were determined: zinc–aluminosilicate s.s. and gahnite. The content of zinc–aluminosilicate s.s. decreases and gahnite increases with the increase of particle size and with temperature. The formation of crystalline phases in the temperature range of the first exothermic peak in dependence of particle size is shown in Fig. 3(a). The amount of the phases is given as the intensity ratio of (1 0 1) and (2 2 0) lines for zinc–aluminosilicate s.s. and gahnite, respectively, and (1 1 1) line of CaF_2. The formation of gahnite on expense of zinc–aluminosilicate s.s. structure during the second exothermic peak is given in Fig. 3(b). Fig. 4 shows XRD patterns of crystalline phases formed during DSC run of glass with particles <63 μm. The temperatures at which the glasses were quenched and subsequently submitted to XRD analysis are given in the figure. At 1143 K, zinc–aluminosilicate s.s. is unambiguously determined as a dominant phase, whereas, the initial crystallization of gahnite is only anticipated by two very broad hump marked in the figure by arrows. The amount of both phases increases with temperature up to 1193 K. However, above this temperature gahnite further increases, while the zinc–aluminosilicate s.s. starts to decrease. At 1423 K, a small amount of zinc–aluminosilicate s.s. and maximum amount of gahnite were observed. (Very small amount of corundum, marked in figure by stars, was also detected at 1193 K and 1293 K). 2θ° positions of XRD lines for zinc–aluminosilicate s.s. that crystallizes at 1143 K do not fit precisely to the $Zn_{0.75}Al_{1.5}Si_{1.5}O_6$ composition cited in JCPDS-ICDD file No. 32-1455 [12]. (To the best of our knowledge we could not find any other zinc–aluminosilicate phase with space group $P6_222$ in JCPDS-ICDD data). Besides, the temperature increase results in a more pronounced displacement of zinc–aluminosilicate lines towards higher 2θ° values. Therefore, to ascertain the change of phase composition with temperature, Rietveld refinement was performed on XRD patterns of the glass powder with particles <63 μm, quenched in DSC apparatus at four different temperatures (1143, 1193, 1293 and 1423 K). Zinc–aluminosilicate s.s. belongs to structures called 'stuffed derivatives of high-quartz'. In these phases the quartz structure is modified by partial replacement of Si^{4+} by Al^{3+} atoms and at the same time, cations of small radius such as Li^+, Mg^{2+} and Zn^{2+} enter normally empty interstitial holes in the silica framework. There is no data for zinc-stuffed derivative of high-quartz, as mentioned above, therefore, we used lithium stuffed derivative of high-quartz ($LiAlSi_2O_6$) derived by Mueller et al. [14] as a model structure, only we put Zn^{2+} cations in tetra-

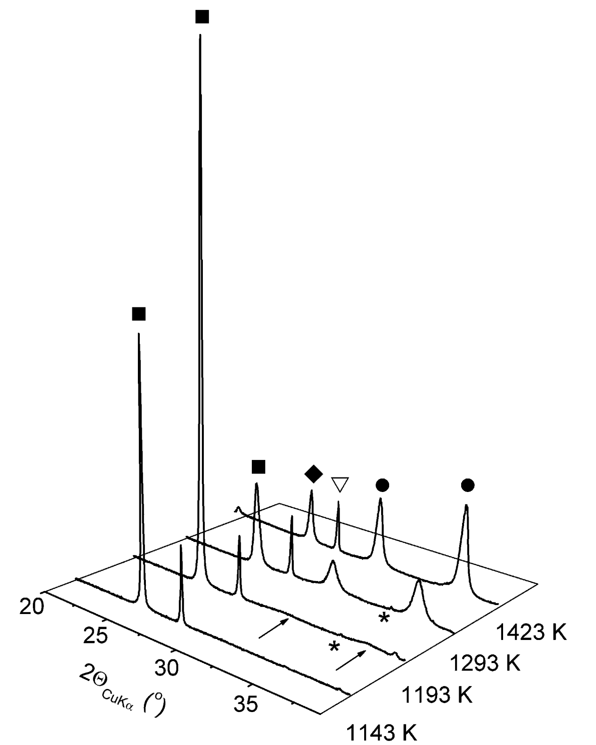

Fig. 4. XRD patterns of glass with particles <63 μm quenched at temperatures marked on y axis in the figure. -■- $Zn_{x/2}Al_xS_{3-x}O_6$, -●- $ZnAl_2O_4$, -Λ- high-quartz, -▽- fluorite and -*- α- Al_2O_3.

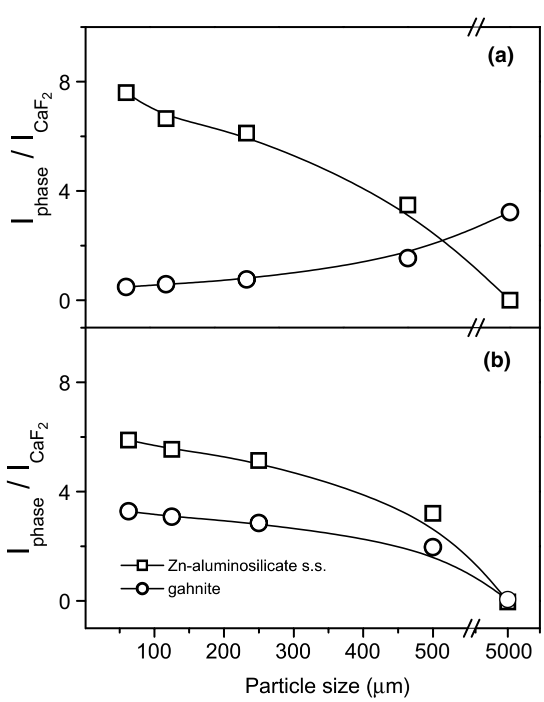

Fig. 3. The content of zinc–aluminosilicate s.s. and gahnite in quenched glasses presented as the intensity ratio of (1 0 1) and (2 2 0) lines for zinc–aluminosilicate s.s. and gahnite, respectively, and (1 1 1) line of CaF_2. The connecting lines are introduced as guidelines for the eye.

hedral 3a positions instead of lithium. Fig. 5(a) shows a plot of the profile matching the result of Rietveld refinement performed for sample quenched at 1193 K. Structure model with zinc in octahedral 3b positions, similar to magnesium stuffed derivative of high-quartz ($MgAl_2Si_2O_6$) was also fitted to experimental XRD pattern and the Rietveld refinement is shown in Fig. 5(b).

The unit cell parameters and the composition of zinc–aluminosilicate s.s., as well as of gahnite are given in Table 2. Literature data for $Zn_{0.75}Al_{1.5}Si_{1.5}O_6$ ($ZnAl_2Si_2O_8$) and gahnite are given underneath the table. Table 3 displays quantitative phase analyses determined on XRD patterns with 10 wt% of CaF_2, which was added to the quenched samples as an internal standard. Since

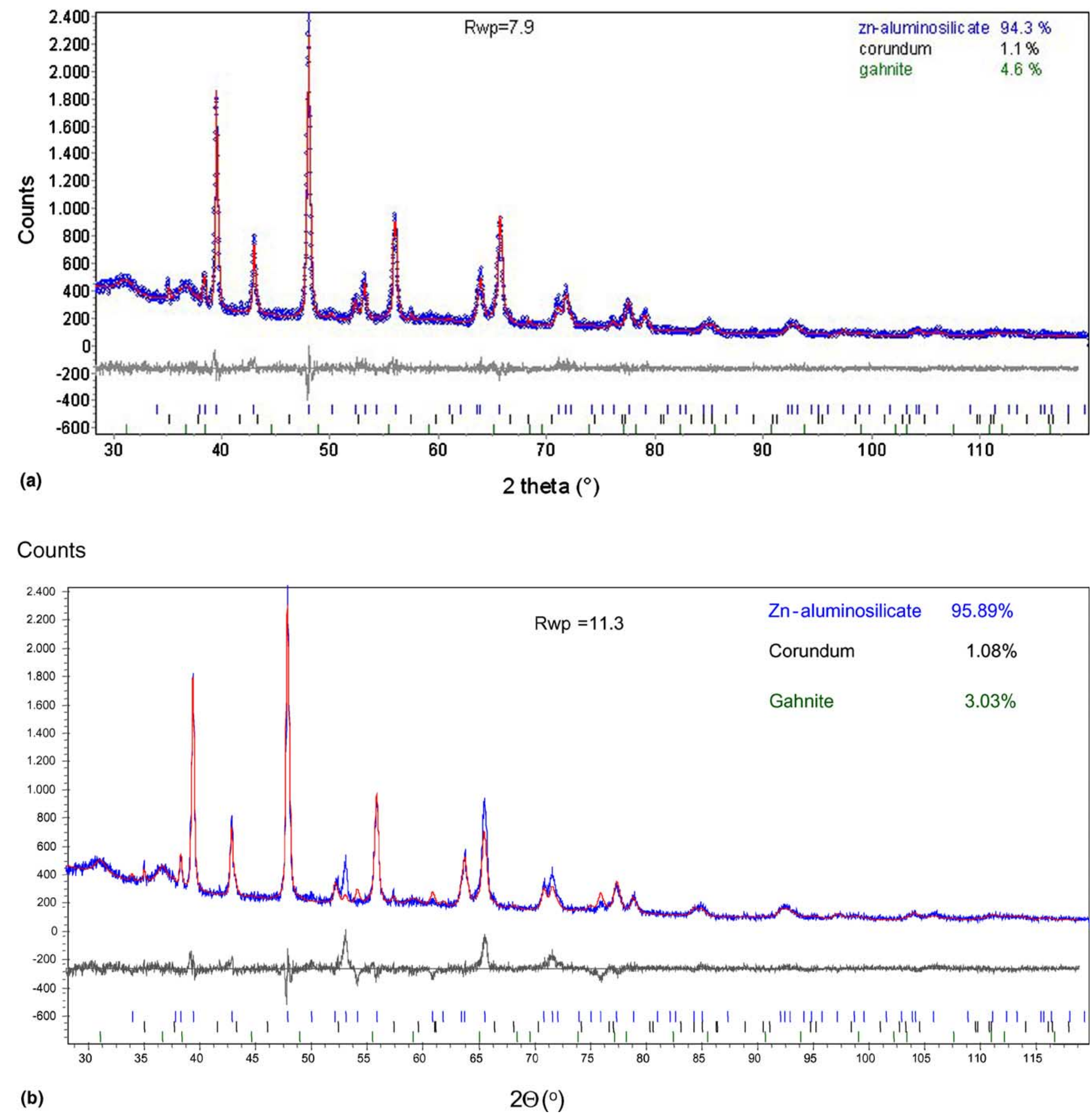

Fig. 5. (a) observed (dots) and calculated intensity (line) profiles for XRD pattern of glass powder (particle size <63 μm) quenched in DSC apparatus at 1293 K, with Zn^{2+} in tetrahedral (3a) position. (b) observed and calculated intensity profiles for the same XRD pattern with Zn^{2+} located in octahedral (3b) position. The short vertical lines indicate the angular position of the allowed Bragg reflections for all refined phases. The lower curve is the difference between the observed and the calculated intensity at each step.

Table 2
Unit cell parameters and the composition of zinc–aluminosilicate s.s. and gahnite obtained by Rietveld refinement of XRD patterns for the heat treated glass powder with particles <63 μm

Phase	Temperature (K)			
	1143	1193	1293	1423
$Zn_{x/2}Al_xSi_{3-x}O_6$[a]				
a (nm)	0.52635(3)	0.52562(2)	0.51909(1)	0.50105(7)
c (nm)	0.54561(4)	0.54546(3)	0.54247(2)	0.54317(1)
Composition	$Zn_{0.69}Al_{1.38}Si_{1.62}O_6$	$Zn_{0.63}Al_{1.27}Si_{1.73}O_6$	$Zn_{0.48}Al_{0.98}Si_{2.03}O_6$	High-quartz[b]
R_{Bragg} (%)	4.17	3.51	3.48	3.34
Gahnite[c]				
a (nm)	n.d.[d]	0.8103(4)	0.8088(3)	0.8087(2)
Composition	–	$ZnAl_2O_4$	$Zn(Al_{0.98}Zn_{0.02})_2O_4$	$Zn(Al_{0.97}Zn_{0.03})_2O_4$
R_{Bragg} (%)	–	5.09	1.85	1.25
R_{wp} (%)	8.25	7.85	7.96	7.52
$R_{wp\ expected}$ (%)	7.19	6.79	6.62	6.44
Goff	1.14	1.15	1.20	1.16

$R_{wp}R_{Bragg}$ and Goff (goodness of fit) are reliability factors which define the quality of Rietveld refinements [9].
[a] JCPDS file 32-1455; $x = 1.5$; $Zn_{0.75}Al_{1.5}Si_{1.5}O_6$; $a = 0.5317$ nm, $c = 0.5458$ nm.
[b] The composition is discussed in the text.
[c] JCPDS 5-669, $ZnAl_2O_4$; $a = 0.8084$ nm.
[d] n.d = not determinable (at the sensitivity limit of the technique).

Table 3
Quantitative phase analyses of heat-treated samples of the glass with particles <63 μm

Temperature of heat-treatment (K)	Composition (wt%)				
	$Zn_{x/2}Al_xSi_{3-x}O_6$	High-quartz	Gahnite	Corundum	Amorphous phase
1143	44	–	traces	–	56
1193	94.3	–	4.6	1.1	n.d.
1293	40.8	–	47.3	6.9	~5
1423		19.9	67.6	–	12.5

n.d. = not determinable.

the quenched samples contain always some amount of amorphous phase and the Rietveld method does not take it into the consideration, the quantitative phase composition was deduced from the ratio of CaF_2 added to a sample and found by the Rietveld refinement. The amount of zinc and aluminum incorporated in high-quartz structure decrease with temperature, which brings about the decrease of unit cell parameters as shown in Fig. 6.

Activation energy for crystallization of zinc–aluminosilicate s.s. in the glass powder with particles <63 μm (p_1 peak on DSC scan (a) in Fig. 1) and for gahnite crystallization (p_2 peak) were evaluated from plots of $\ln(\beta/T_p^2)$ versus $1/T_p$ and the obtained results are given in Fig. 7.

TEM micrograph of bulk glass nucleated at 1023 K for 2 h, and corresponding SAD pattern (as an insert) are given in Fig. 8. In Fig. 9, the microstructure (with corresponding SAD pattern given as an insert) and microanalyses of crystalline and amorphous regions are correlated. The same correlation is made for the translucent glass-ceramic obtained by heat-treatment at 1423 K in Fig. 10.

4. Discussion

4.1. DSC and XRD analyses

DSC scan (e) in Fig. 1 and XRD pattern (~5 mm) in Fig. 2 show that in bulk sample only gahnite, $ZnAl_2O_4$, crystallizes. However, in the glass powders (particles <500 μm, scans (a)–(d)) two crystalline phases occur: zinc–aluminosilicate solid solution, $Zn_{x/2}Al_xSi_{3-x}O_6$, and gahnite, $ZnAl_2O_4$. Zinc–aluminosilicate s.s. with high-quartz structure (zinc-stuffed derivative of high-quartz) is metastable phase, and crystallizes only during the appearance of the first exothermic peak on DSC scans. Gahnite on the other hand, crystallizes during the first exothermic peak simultaneously with zinc–aluminosilicate s.s., but more pronounced during the second exothermic event. The crystallization of metastable zinc–aluminosilicate s.s. occurs on the surface of particles, whereas gahnite crystallizes by volume crystallization [7]. Consequently, with the increase of particles size, the amount of the zinc–aluminosilicate s.s. decreases and the amount of gahnite increases

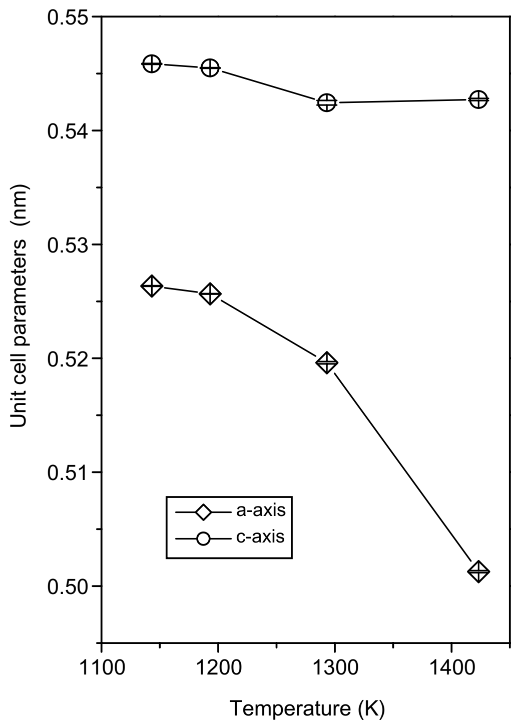

Fig. 6. The unit cell parameters of zinc–aluminosilicate vs. temperature. Standard deviations of the parameters are within the marked values. The lines are introduced as guidelines for the eye.

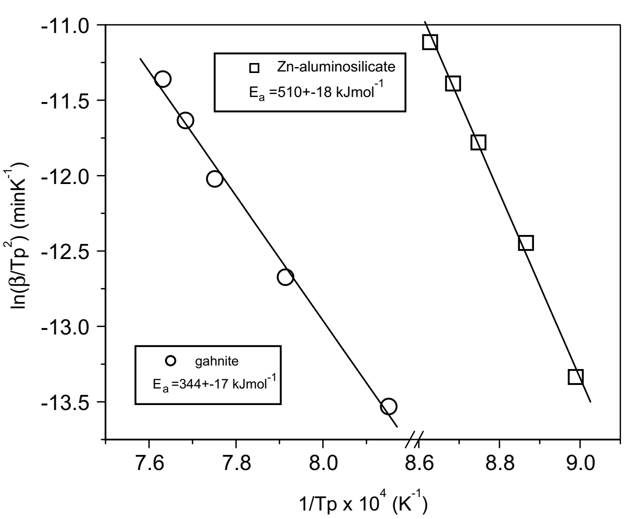

Fig. 7. Kissinger plots (Eq. (1)) for first exothermic peak (attributed to crystallization of zinc–aluminosilicates.s.), and second exothermic peak (attributed to gahnite formation). Particle size <63 μm.

(Fig. 3(a)). The amount of gahnite, which crystallizes during the second peak is mutually dependent on zinc–aluminosilicate s.s. Accordingly, its amount decreases with the decrease of zinc–aluminosilicate s.s. (Fig. 3(b)). Eventually, in bulk glass only gahnite crystallizes. Considering the surface crystallization, an increase in the specific surface area of sample acts similarly as a nucleating agent, therefore, as the specific surface area

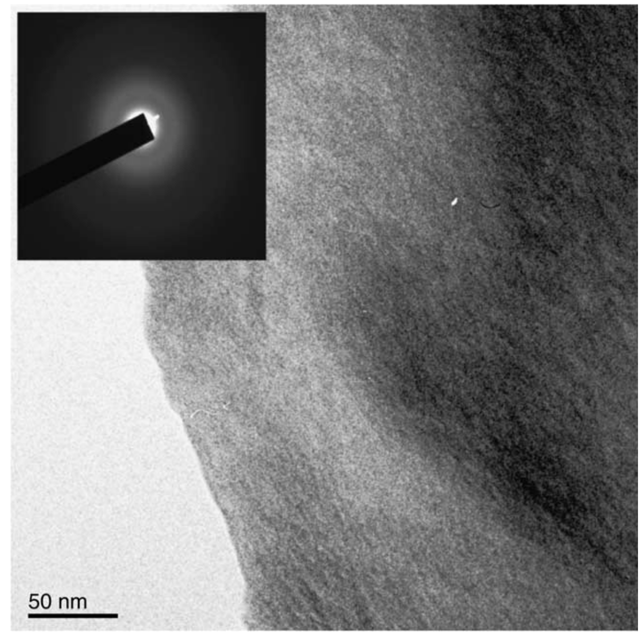

Fig. 8. TEM micrograph of bulk glass nucleated at 1023 K. SAD pattern is given as an insert in the figure.

increases, i.e. as the particle size decreases, the position of the exothermic peak is shifted to lower temperatures (first peak in Fig. 1). On the contrary, for volume crystallization the effect of specific surface area on the shift of the peak is either small or does not appear at all. Therefore, there is no shift of the second exothermic peak on DSC scans in Fig. 1. From chemical analysis of the glass (Table 1), the crystallization of metastable high-quartz structure was expected. According to Petzoldt [6] the crystallization of zinc–aluminosilicate s.s. is to occur, if the composition of glass lies on the $ZnAl_2O_4$–SiO_2 connecting line in ZnO–Al_2O_3–SiO_2 system with more than 50 mol% SiO_2.

The composition of glass-ceramics in dependence of temperature is studied only for the glass powder with particles <63 μm. The glass was quenched at 1143, 1193, 1293 and 1423 K. There is some shift of diffraction lines of zinc–aluminosilicate s.s. at 1143 K in comparison to the phase $Zn_{0.75}Al_{1.5}Si_{1.5}O_6$ ($ZnAl_2Si_2O_8$) cited in JCPDS-ICDD File No. 32-1455 [12] (Fig. 4), as was mentioned above. The experimental and calculated X-ray profiles are plotted in Fig. 5. As shown (Fig. 5a), there exists an excellent agreement between observed and calculated profiles of zinc–aluminosilicate phase, if Zn^{2+} was located in tetrahedral (3a) positions. On the contrary, there exists a great discrepancy between calculated and experimental profiles if Zn^{2+} was located in octahedral positions. Gahnite was simultaneously refined with zinc–aluminosilicate.

The composition $Zn_{0.69}Al_{1.38}Si_{1.62}O_6$ richer in silica than reported in [12] was ascertained at 1143 K (Table 2). This difference could be explained by different

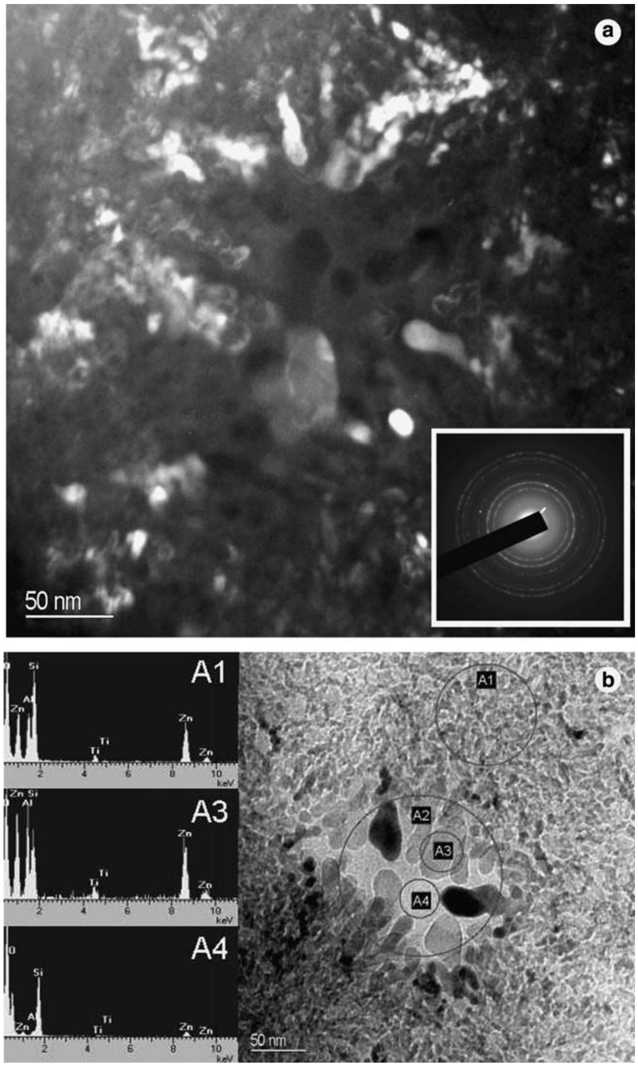

Fig. 9. TEM micrographs of bulk glass heat-treated at 1156 K: (a) dark field and SAD pattern given as an insert. (b) bright field and EDX analyses of crystalline region A1, the grain A3 and amorphous part of the sample (A4).

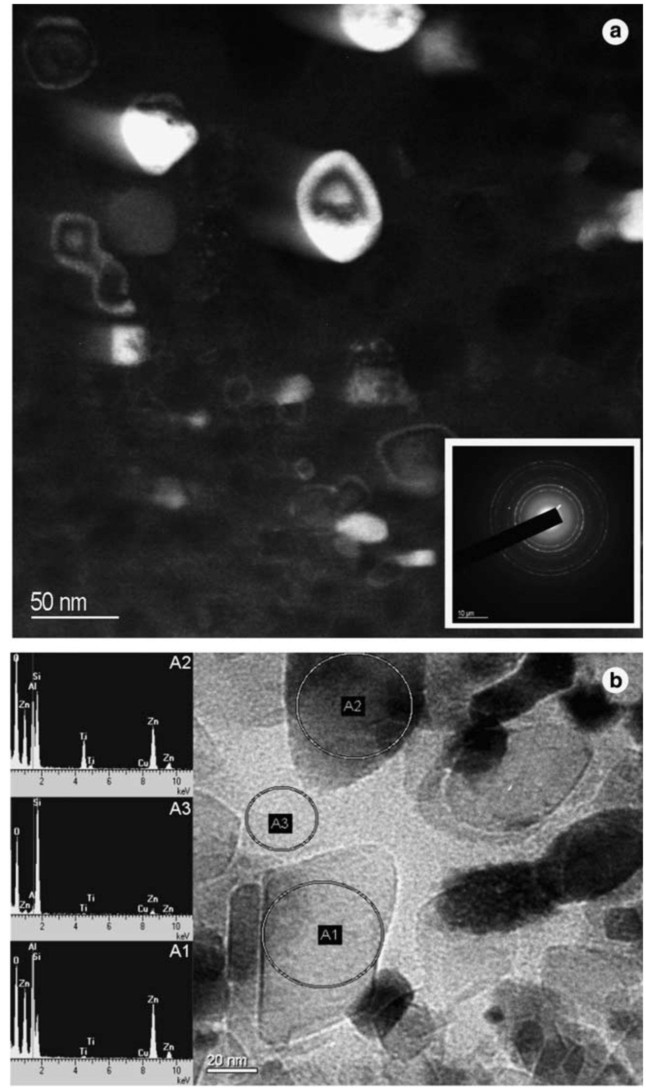

Fig. 10. TEM micrographs of bulk glass heat-treated at 1423 K: (a) dark field and SAD pattern given as an insert. (b) bright field and EDX analyses of A1 and A2 grains and amorphous part of the sample (A3).

technique used at the sample preparation. Roye et al. [12] devitrified the glass at 1158 K that possessed the same Zn:Al:Si = 0.5:1:1 ratio as the crystalline $Zn_{0.75}Al_{1.5}Si_{1.5}O_6$ phase. In our work the ratio was Zn:Al:Si = 0.32:0.82:1. Negligible change in composition of zinc–aluminosilicate in the temperature range of the first exothermic peak (between 1143 and 1193 K) was observed, while the amount of the phase increases and attains more than 90 wt% at 1193 K (Table 3). Its amount started to decrease in larger extent above 1193 K, with simultaneous decrease of unit cell parameters (Fig. 6). This is due to the exsolution of zinc and aluminum from the metastable high-quartz structure followed by the crystallization of gahnite and the increase of amorphous SiO_2 phase in the system. At 1423 K, the best fitting result is obtained using the parameters of even pure high-quartz structure (Table 2). However, it should be assumed that the presence of very small (not determinable) amounts of Al^{3+} and Zn^{2+} stabilize the high-temperature structure of quartz, which is otherwise unstable at room temperature. Matsumoto et al. [16] studying the formation of high-quartz solid solution from Zn-exchanged Zeolite A came to the same conclusion. There is no significant difference in *a* parameter of gahnite at 1293 and 1423 K, (Table 2). Some incorporation of aluminum in tetrahedral sites and zinc in octahedral sites in spinel are always possible.

4.2. Determination of activation energy

The activation energy of gahnite crystallization $344 \pm 17\,kJ\,mol^{-1}$ (Fig. 7) matches well with the values for crystal growth of gahnite in similar glasses cited in literature [2,17], although no nucleation of the sample was undertaken. This can be attributed to the fact that the high-quartz structure acts as precursor for gahnite [7]. According to our knowledge the activation energy for crystallization of metastable zinc–aluminosilicate was not yet determined. The amount of gahnite that crystallizes during the first peak in the glass with particles <63 μm is small and can be neglected; therefore, the determined value of $510 \pm 18\,kJ\,mol^{-1}$ has to be attributed solely to activation energy for crystallization of zinc–aluminosilicate s.s.

4.3. TEM analyses

Glass nucleated at 1023 K for 2 h is still amorphous and homogeneous as shown by TEM micrograph and electron diffraction in Fig. 8. (SAD is given as an insert in the figure). No phase separation by TEM is noticeable. Nanosized (<20 nm) gahnite crystals are observed in the glass heat-treated at 1156 K (Fig. 9), and the obtained glass-ceramic is still transparent. By heat-treatment at higher temperature (1423 K) gahnite grains increased to about 30–40 nm and glass becomes translucent (Fig. 10). EDX analyses (inserts in Figs. 9(b) and 10(b)) confirm enrichment of titanium in alumina-rich crystalline area rather than in silica-rich amorphous glass area. According to Beall and Pickney [3] TiO_2 serves as an intergral component of spinel structure with Ti^{4+} entering the octahedral (b) site in spinel structure.

5. Conclusion

- Nanocrystalline transparent glass-ceramics with gahnite as the single phase could be achieved by controlled heat treatment of bulk glasses with composition of 55.6 mol% SiO_2, 22.8 mol%Al_2O_3, 17.7 mol% ZnO and 3.84 mol% of TiO_2 as nucleating agent.
- Metastable zinc–aluminosilicate s.s. preceded the formation of gahnite in glasses powders. The crystallization of metastable zinc–aluminosilicate s.s. occurred on the surface of particles, whereas gahnite crystallized by volume crystallization mechanism.
- The composition of initially formed zinc–aluminosilicate s.s. determined by Rietveld refinement of XRD patterns has been $Zn_{0.69}Al_{1.38}Si_{1.62}O_6$. With the temperature increase, the amount of zinc and aluminum incorporated in the structure of high-quartz decreased and eventually almost pure high-quartz structure was formed at 1423 K. Simultaneously, crystallization of gahnite and enrichment of samples in amorphous SiO_2-rich glassy phase occurred.
- The activation energy of gahnite crystallization $344 \pm 17\,kJmol^{-1}$ matches well with the values for crystal growth of gahnite in similar glasses cited in literature, although no nucleation of the sample was undertaken. That is attributed to the fact that the high-quartz structure acts as precursor for gahnite.

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